small number of other ones. This network forms the skeleton on which space itself is constructed.

That's a start, but we want to do even more. When someone asks you how far apart two different cities are, they're looking for something a bit more specific than "near" or "far." They want an actual distance, and that's what the metric on spacetime ordinarily lets us calculate. In our abstract wave function divided up into degrees of freedom, we haven't yet constructed a full geometry, just a notion of near and far.

We can do better. Remember the intuition from vacuum states in quantum field theory that Jacobson used to derive Einstein's equation: the entanglement entropy of a region of space is proportional to the area of its boundary. In our current context of a quantum state described in terms of abstract degrees of freedom, we don't know what "area" is supposed to mean. But we do have entanglement between the degrees of freedom, and for any collection of them we can compute their entropy.

So once again following our reverse-engineering philosophy, we can *define* the "area" of a collection of degrees of freedom to be proportional to its entanglement entropy. In fact, we can assert this for every possible subset of degrees of freedom, assigning areas to every surface we can imagine drawing within our network. Happily, mathematicians long ago figured out that knowing the area of every possible surface in a region is enough to fully determine the geometry of that region; it's completely equivalent to knowing the metric everywhere. In other words, the combination of (1) knowing how our degrees of freedom are entangled, and (2) postulating that the entropy of any collection of degrees of freedom defines an area of the boundary around that collection, suffices to fully determine the geometry of our emergent space.

